Decimals & Money

The mission where being wrong costs real money.

THE BIG QUESTION

A decimal and a fraction can be the same amount — so why do we have both?

HANDS-ON PROJECT

The $40 Grocery Mission

Plan three real dinners on a real $40 budget, shop it, then reconcile the receipt against the plan.

FLUENCY GAME

Change Sprint

Make an exact amount with the fewest coins. Beat your own record.

AXIOM · MISSION COMMANDER, NINE TOURS

"Brock — the receipt doesn't care how you feel about your answer. That's what makes it the best marker you'll ever have."

NAME ______________

DATE ______________

Pre-Assessment

Twelve items, six strands. Anything you already own, we skip.

This is not a test. Nothing here counts for a grade. Write dollar signs and decimal points properly — that's part of what's being checked.

STRAND A
READING DECIMALS

A1 0.3 = _____ / 10

A2 0.07 = _____ / 100

STRAND B
COMPARING

B1 Circle the bigger: 0.4 0.35

B2 Order smallest first: 0.5, 0.45, 0.055

STRAND C
ADDING MONEY

C1 $3.45 + $2.30 = _______

C2 3.4 + 0.75 = _______

STRAND D
MAKING CHANGE

D1 $20.00 – $13.75 = _______

D2 Fewest coins to make 67¢? List them.

STRAND E
ROUNDING

E1 Round $23.49 to the nearest dollar.

E2 Estimate $3.95 + $6.10 without adding exactly.

STRAND F
BUDGETING

F1 You have $40 and spend $27.85. How much is left?

F2 Cheaper: 4 for $6.00 or $1.60 each? Show why.

Skip Chart

Score the pre-assessment, then cross days off the four-week calendar.

2 / 2 → SKIP
1 / 2 → COMPRESS
0 / 2 → TEACH

STRAND	ITEMS	COVERS	SCORE	IF 2/2, REMOVE
A · Reading decimals	A1 A2	Week 1 · lesson 1.1	___ / 2	Day 1.1. Warm-Up tier drops to 3 items all unit.
B · Comparing	B1 B2	Week 1 · lessons 1.2–1.3	___ / 2	Days 1.2 and 1.3. B1 wrong is diagnostic — do not skip on B2 alone.
C · Adding money	C1 C2	Week 2 · lessons 2.1–2.2	___ / 2	Days 2.1–2.2. C2 is the real test; C1 alone proves nothing.
D · Making change	D1 D2	Week 1 · 1.4 and Week 2 · 2.3	___ / 2	Day 2.3. Keep the Change Sprint — it's fast and he'll like it.
E · Rounding	E1 E2	Week 3 · lessons 3.1 and 2.4	___ / 2	Day 3.1 and day 2.4.
F · Budgeting	F1 F2	Week 3 · 3.2–3.4	___ / 2	Days 3.2–3.3. Never skip Week 4 — the project is the point of the mission.

Where the saved time goes

Into the grocery mission. A second shop, a bigger budget, or a comparison between two stores all beat another worksheet. Write the reallocation here:

....

....

Revised Mission 05 plan

WEEK 1 ☐ full ☐ compressed ☐ skip
WEEK 2 ☐ full ☐ compressed ☐ skip
WEEK 3 ☐ full ☐ compressed ☐ skip
WEEK 4 ☐ full ☐ compressed ☐ skip

NAME ___________

DATE ___________

Tenths and Hundredths

A decimal is a fraction whose bottom number is always ten or a hundred.

Warm-Up

WRITE EACH ONE AS A FRACTION

0.3	0.7	0.09	0.5	1/2	1/4
____	____	____	____	____	____

SHADE 0.35 ON THE HUNDRED GRID

Shade thirty-five squares. Then write the same amount three ways:

as a fraction ....

in words ....

as money ....

Core

1. 0.6 = ____ hundredths ....
2. $2.35 = ____ cents ....
3. $0.08 = ____ cents ....
4. 3/10 + 4/10 = ____ ....
5. 0.25 + 0.25 Answer in hundredths, then as a fraction in simplest form: ....

Challenge

NOT GRADED

C1. Which is bigger, 0.7 or 0.65? Explain why the one with fewer digits wins.

....

C2. How many hundredths are in one whole? How many in two wholes?

....

C3. Write 1/3 as a decimal. What goes wrong, and what does that tell you about why we keep fractions?

....

NAME ___________

DATE ___________

Decimals on the Line

More digits does not mean bigger. That's the whole lesson.

Warm-Up

WRITE EACH IN HUNDREDTHS

0.4	0.9	0.05	0.75	0.1	1.0

MARK 0.2, 0.65, 0.9 AND 0.35 ON THIS LINE

0 ———————————————— 1

0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9

Core

1. 0.65 sits between which two tenths?
2. What decimal sits halfway between 0.2 and 0.3?
3. Write 0.8 as a number of tenths.
4. Which is bigger, 0.4 or 0.35? Write both as hundredths first.
5. 0.3 + 0.45 = _______

Challenge

C1. Order 0.5, 0.45 and 0.055 smallest first. Writing them all as thousandths makes it easy — do that first.

C2. 0.9 – 0.35 = ______

C3. Name a decimal between 0.6 and 0.61. Then name one between that and 0.61 again.

NAME ___________

DATE ___________

Money Math

Line up the decimal points, not the right-hand ends.

Warm-Up

WRITE EACH AMOUNT IN CENTS

$1.50	$0.75	$3.05	$10.00	$0.99	$2.20
____	____	____	____	____	____

THE MOVE · COUNT UP, DON'T BORROW

$13.75 + 25¢ → $14.00

$14.00 + $6 → $20.00

change = $6.25

No crossed-out zeros, no borrowing across three columns. This is how every shopkeeper on earth does it, and it is faster than the algorithm.

Core

1. $3.45 + $2.30 ....

2. $20.00 – $13.75 ....

3. $4.99 + $3.99 ....

4. Three items at $2.50 each ....

5. $10 – ($2.50 + $3.75) Total the two prices first, then count up from there: ....

Challenge

C1. Which is cheaper per item: 4 for $6.00, or $1.60 each? Prove it.

....

C2. $20 split evenly between 8 people. How much each?

....

C3. You pay for a $7.30 item with a $10 note and get $3.70 back. Is that right? Say how you checked.

....

NAME ___________

DATE ___________

Rounding & Estimating

Round before you compute, so you know what the answer should look like.

Warm-Up

ROUND TO THE NEAREST DOLLAR

$3.45	$7.80	$12.50	$0.60	$9.49	$5.55
____	____	____	____	____	____

ESTIMATE, THEN COMPUTE

$3.95 + $6.10 → about $4 + $6 = **$10**

exact: **$10.05** — the estimate was five cents low

If the exact answer lands far from the estimate, one of them is wrong. Nine times out of ten it's the exact one.

Core

1. Round $4.678 to the nearest cent. ....

2. Round $23.49 to the nearest dollar. ....

3. Estimate $3.95 + $6.10 to the nearest dollar. ....

4. Now compute $3.95 + $6.10 exactly. ....

5. Round 0.482 to the nearest hundredth. ....

Challenge

C1. Estimate $19.99 × 3, then compute it exactly. Why is the trick with $20 so much easier?

....

C2. You buy two things at $19.99 and pay with $50. What's your change?

....

C3. A shop rounds every price up to the nearest dollar. Whose side is that rounding on, and by how much on a $27.10 basket?

....

The $40 Mission Begins

Plan on paper today. The shopping happens in Week 4.

NAME ___________

DATE ___________

Three dinners, forty dollars

DINNER	WHAT'S IN IT	ESTIMATED
1		
2		
3		
Estimated total		
Under or over $40, and by how much?		

★ Budget arithmetic

- Budget $40, spent $27.85. Left? ....
- $12.40 + $9.85 + $15.30 = ? ....
- Is that total under $40? By how much? ....
- $1.29 × 4 = ? ....
- $5.00 shared 4 ways ....
- Cheaper per ounce: 12 oz for $3.60 or 16 oz for $4.00? ....

Change Sprint

Make each amount with the fewest coins possible. Quarters, dimes, nickels and pennies only. Write the coins, then the count.

67¢ .... 41¢ .... 99¢ .... 36¢ ....

The real question

Which amount under a dollar needs the most coins to make? Find it, and explain why that particular amount is the worst one.

....

....

The Rest of the Mission

Three weeks left, and the last one leaves the house. Same three-tier shape every day.

Week 2 · Adding & Subtracting

LINE UP THE POINTS

2.1 Mon	2.2 Tue	2.3 Wed	2.4 Thu	Fri
Line up the decimal points, not the right-hand ends.	Holding places with zeros: 3.4 becomes 3.40.	Making change from $20 by counting up.	Estimate first, to the nearest dollar.	Change Sprint — fewest coins, against the clock.

Week 3 · Rounding & Money Math

MID-UNIT QUIZ FRIDAY · 85% TO ADVANCE

3.1 Mon	3.2 Tue	3.3 Wed	3.4 Thu	Fri
Rounding to the nearest cent, and to the nearest dollar.	Unit price. Divide, then compare.	Is it a deal? Three for $5 against $1.75 each.	Multi-step money with change from a twenty.	Mid-unit quiz, 8 items, Weeks 1–3.

Week 4 · Budget & Project

UNIT TEST FRIDAY · BADGE AWARDED

4.1 Mon	4.2 Tue	4.3 Wed	Thu	Fri
Build the budget: three dinners, $40 ceiling, prices estimated.	Shop it. Real store, running total kept by hand.	Reconcile the receipt. Where was the estimate wrong?	Error journal sweep.	Mission 05 test: 12 items + one explanation.

MATERIALS, WHOLE UNIT

Real coins and notes · a $40 float for the shop · store flyers or a grocery app · a clipboard for the running total · the error journal

THE SHOP TRIP

He keeps the running total, not you. If it goes over $40 at the till, that is the lesson — do not quietly cover it.

THE MOST COMMON STALL

Adding 3.4 + 0.75 as if it were 3.4 + 0.7 + 5. Write the zero: 3.40. Every time, until he stops needing to.

Mid-Unit Quiz

8 items · Weeks 1–3 · 7 of 8 keeps you flying

NAME ___________

DATE ___________

___ / 8

1. 0.6 = _____ / 100

2. Which is bigger, 0.4 or 0.35? Show why.

3. Mark 0.25, 0.6 and 0.85 on this line.

0 ———————————————— 1

4. 3.4 + 0.75 = _______

5. $20.00 – $13.75 = _______

6. Round $23.49 to the nearest dollar.

7. Cheaper per item: 4 for $6.00 or $1.60 each?

CHALLENGE ITEM

8. You buy items at $4.25, $3.80 and $6.99, and pay with a $20 note. What is your change? Estimate first.

NAME ___________

DATE ___________

Mission 05 Test

12 items + one explanation · part 1 of 2

1. 0.9 = ____/10

2. 0.07 = ____/100

3. Write 3/4 as a decimal.

4. Order smallest first: 0.5, 0.45, 0.055

5. 0.9 – 0.35 = _______

6. $4.99 + $3.99

7. 3.4 + 0.75

8. $50 – $39.98

9. Round $4.678 to the nearest cent, and $23.49 to the nearest dollar.

10. Estimate $19.99 × 3, then give the exact answer.

Reasoning & the Big Question

ITEMS ___ / 12
EXPLANATION ___ / 4

11. A 12 oz bag costs $3.60. A 16 oz bag costs $4.00. Which is better value, and by how much per ounce?

12. Budget $40. You spend $12.40, $9.85 and $15.30. Are you under? By how much?

CHALLENGE ITEM

Make 99¢ with the fewest coins. Then explain which amount under a dollar needs the most.

EXPLAIN YOUR THINKING · WORTH 4 POINTS

A decimal and a fraction can be the same amount — so why do we have both?

Give one amount where the decimal is the better tool and one where the fraction is. Then find an amount that has no exact decimal at all, and say what that proves. Spelling doesn't count. Reasoning does.

TROPHY BAND · CIRCLE ONE

85–100%
Mission Commander

Budget Badge earned. Mission 06 opens — and Challenge becomes your Core.

70–84%
Flight Engineer

Two targeted days on the flagged items, then retake those items only.

Below 70%
Ground Crew

Back to the hundred grid and real coins before any more written work.

Key A · Pre-Assessment, Lessons 1.1–1.2

Pre-Assessment

A1 **3/10**   A2 **7/100**

B1 **0.4** (0.40 against 0.35)

B2 **0.055, 0.45, 0.5**

C1 **$5.75**   C2 **4.15**

D1 **$6.25**

D2 25 + 25 + 10 + 5 + 1 + 1 = **6 coins**

E1 **$23**

E2 about **$10** ($4 + $6)

F1 **$12.15**

F2 4 for $6.00 → **$1.50 each**, cheaper

Scoring note. B1 answered as 0.35 means he is counting digits instead of place value. That is the diagnostic miss of this mission — run Week 1 in full whatever else he scored.

Lesson 1.1 · Tenths and Hundredths

Warm-Up 3/10 · 7/10 · 9/100 · 5/10 · 50/100 · 25/100

Grid 35/100 · thirty-five hundredths · $0.35

Core 1. **60**   2. **235**   3. **8**

4. **7/10**   5. **50/100 = 1/2**

Lesson 1.1 Challenge. C1 · 0.7 is bigger; 0.70 against 0.65. The digit count is irrelevant — only the place value matters. C2 · 100 in one whole, 200 in two. C3 · 1/3 = 0.333… and never stops. That is exactly why fractions survive: some amounts have no exact decimal, so the fraction is the only honest way to write them.

Lesson 1.2 · Decimals on the Line

Warm-Up 40 · 90 · 5 · 75 · 10 · 100

Core

1. between **0.6 and 0.7**
2. **0.25**
3. **8 tenths**
4. **0.4** — 0.40 against 0.35
5. **0.75**

Challenge. C1 · 0.055, 0.45, 0.5 — as thousandths that's 55, 450 and 500. C2 · 0.55. C3 · 0.605 works, then 0.6051, and it never runs out. That there is always another decimal in between is worth pausing on; it surprises most adults too.

Watch for: "0.35 is bigger because it has more numbers." Go straight to the hundred grid — shading 40 against 35 ends the argument in ten seconds.

Key B · Lessons 1.3–1.4, Enrichment, Quiz

Lesson 1.3 · Money Math

Warm-Up 150 · 75 · 305 · 1000 · 99 · 220

Core

1. **$5.75**
2. **$6.25**
3. **$8.98**
4. **$7.50**
5. $6.25 spent → **$3.75** change

Challenge. C1 · 4 for $6.00 is $1.50 each, cheaper by 10¢ an item. C2 · $2.50 each. C3 · No — $10.00 – $7.30 is $2.70, so the change was a dollar too much. Accept any check, but counting up from $7.30 is the one to praise.

Lesson 1.4 · Rounding & Estimating

Warm-Up $3 · $8 · $13 · $1 · $9 · $6

1. **$4.68**
2. **$23**
3. about **$10**
4. **$10.05**
5. **0.48**

Challenge. C1 · $60 estimated, **$59.97** exact; $20 × 3 minus 3¢ is far easier than three long multiplications. C2 · $39.98 spent, **$10.02** change. C3 · the rounding is on the shop's side; a $27.10 basket becomes $28, so 90¢ extra. Accept any answer that identifies who gains.

Friday · The $40 Mission

Budget arithmetic

$40 - $27.85 = **$12.15**

$12.40 + $9.85 + $15.30 = **$37.55**

Under by **$2.45**

$1.29 × 4 = **$5.16**

$5.00 ÷ 4 = **$1.25**

16 oz for $4.00 = **25¢/oz**, cheaper

Change Sprint. 67¢ → 25, 25, 10, 5, 1, 1 (six coins). 41¢ → 25, 10, 5, 1 (four). 99¢ → 25, 25, 25, 10, 10, 1, 1, 1, 1 (nine). 36¢ → 25, 10, 1 (three).

The real question. 99¢ needs the most, at nine coins — it is one cent short of a dollar in every denomination at once, so every coin size fails you by a hair. Accept any answer that identifies 99¢ and explains the near-miss.

Mid-Unit Quiz · out of 8

1. **60**/100   2. **0.4** (0.40 > 0.35)

3. marks at a quarter, six tenths, 85 hundredths

4. **4.15**   5. **$6.25**

6. **$23**   7. 4 for $6.00 → **$1.50 each**

8. $15.04 spent → **$4.96** change

7/8 or better → continue. A miss on 4 is the missing zero, not the arithmetic. A miss on 2 sends you back to the hundred grid before Week 4's shopping trip.

Key C · Mission 05 Test & Scoring

Items 1–12 · one point each

1. **9/10**
2. **7/100**
3. **0.75**
4. **0.055, 0.45, 0.5**
5. **0.55**
6. **$8.98**
7. **4.15**
8. **$10.02**
9. **$4.68** and **$23**
10. estimate **$60**, exact **$59.97**
11. 16 oz — **25¢/oz against 30¢/oz**
12. total **$37.55**, under by **$2.45**

Challenge item. 99¢ takes nine coins; it is the worst amount under a dollar because it misses every denomination by one cent.

Explanation · 4 points

Score the reasoning, not the writing. Award one point each:

☐ 1 · Gives an amount where the decimal is better, with a reason

☐ 2 · Gives an amount where the fraction is better, with a reason

☐ 3 · Finds an amount with no exact decimal (1/3 is the usual one)

☐ 4 · Concludes that neither form replaces the other

Model answer, kid-sized. "Money is easier as decimals because everything lines up in columns. Sharing a pizza three ways is easier as 1/3, because 0.333... goes on forever and never lands. So you keep both — they're good at different jobs."

Total 16 points. 14+ Mission Commander · 11–13 Flight Engineer · 10 or fewer Ground Crew. A miss on item 7 alone is the missing zero and never blocks the badge; a miss on 11 and 12 together means the project needs repeating with a different budget.

Error Journal · photocopy this page

One row per mistake, for the whole mission. The "why" line must name a step. "I was careless" doesn't count.

DATE	PROBLEM	I GOT / ANSWER	WHY IT HAPPENED — ONE SENTENCE